

User Guidelines - Quick Start Guide

1. What This Project Does

Audio Frame Identifier finds the **first moment someone speaks a specific sentence** in a video, and saves a picture (frame) of the video at that exact moment.

- It listens to the **spoken audio**, not subtitles or on-screen text.
- It converts speech to text (transcription), searches for your sentence/dialogue, finds when it was said for the first time, and saves the matching video frame as an image.

User provides:

1. A video URL
2. The sentence/dialogue you're looking for (as spoken)
3. A choice of approach: **1 (Full transcription)** or **2 (Progressive transcription)**

The program gives you back:

- Whether the sentence was found
- The start/end time it was spoken
- The recognized text
- A similarity score (how close the recognized text is to what you typed)
- The saved frame's file location

2. Before You Start (Prerequisites)

- Python 3.11 (or a similarly configured environment)
- FFmpeg and FFprobe (tools for handling video/audio)
- Project dependencies installed from `requirements.txt`
- An active internet connection (needed to download videos)
- A valid, supported video URL
- Knowledge of the exact sentence spoken in the video

Set up the environment:

Install python: winget install Python.Python.3.11

Verify python: python --version

Install FFmpeg , FFprobe: winget install Gyan.FFmpeg.Shared

Verify: ffmpeg -version, ffprobe -version

Install dependencies

Make sure the virtual environment is activated, then run:

```
python -m pip install -r requirements.txt
```

Verify the installations:

Verify yt-dlp: python -m yt_dlp --version

Verify Faster-Whisper: python -c "from faster_whisper import WhisperModel; print('faster-whisper OK')"

Verify pytest: python -m pytest --version

3. Running the Program

Open a terminal (PowerShell on Windows) and follow these steps exactly:

cd <path where the downloaded/cloned project is present>	:Enter into your project
.env\Scripts\Activate.ps1	:Activate virtual environment
python main.py	:Run the program

The program will then ask you three questions, one at a time:

Enter video URL:

Enter dialogue:

Choose approach:

1. Full transcription
2. Progressive transcription

Enter choice (1/2):

After you answer, the program automatically:

1. Downloads the video
2. Transcribes the speech
3. Searches for your sentence
4. Extracts the matching frame
5. Reports the result (time stamp, frame number,font/dialogue, path to frame image)

4. Understanding Inputs

Video URL

The downloader (yt-dlp) is built to support many online video sources, not just YouTube. It has been tested with YouTube and a direct Wikimedia Commons WebM link. Not every URL on the internet is guaranteed to work. The downloaded file keeps its real extension (e.g., .mp4, .webm).

Dialogue (the sentence to be searched for)

Type the sentence as it is **spoken** in the video. It doesn't need to match the AI's recognized text word-for-word as the system uses approximate matching. For example:

Field	Value
What you typed	"sorry sir but you cannot come here"
What the AI heard	"sorry sir but you cannot come in"
Similarity score	0.857

Even with small wording differences, the match succeeds because the similarity is high enough. The system always returns the **first** qualifying occurrence of your sentence.

5. Choosing an Approach

Situation	Recommended approach
Short video	Approach 1 (Full transcription)
Long video, sentence likely near the start	Approach 2 (Progressive transcription)
Repeated searches on the same, already-fully-transcribed video	Approach 1
Long video, unsure where sentence occurs	Approach 2

Approach 1 — Full Transcription

Flow: **Video** → **Audio** → **complete transcription** → **word-level transcript** → **search** → **timestamp** → **frame**

- Good for short videos or when you plan to run multiple searches against the same transcript.
- Downside: it may transcribe the entire video even if your sentence appears near the beginning, which takes longer.

Approach 2 — Progressive Transcription

Works in 30-second chunk. It stops as soon as your sentence is found, saving time on long videos. It also saves its progress, so it can reuse already-transcribed portions later.

6. Where Files Are Stored

Type	Location
Downloaded videos	data/raw/
Extracted audio	data/audio/
Transcripts	outputs/transcripts/
Saved frames	outputs/frames/

These are separate downloading a video does not automatically create its transcript. A transcript can be missing, partial, or complete depending on what you've run.

Downloaded files get unique sequential names, keeping their real extension, e.g.:

```
video_001.mp4
video_002.webm
```

This naming lets the program reuse previously downloaded/processed files instead of redoing work, which saves time.

7. How Matching Works

The program normalizes text (cleans up casing/punctuation) and compares word sequences for similarity. By default, a match needs at least **0.80 (80%) similarity**, and the matched text must start with the same first word as your input to avoid odd partial matches. The **first** result meeting this bar is returned.

8. If the Sentence Isn't Found

Possible reasons:

- The phrase isn't actually spoken in the video
- The AI transcribed it with different wording
- Audio is noisy, quiet, or distorted
- Strong accent or unusual pronunciation
- Overlapping speakers
- The phrase is too specific/long
- A processing issue occurred

What to try:

- Double-check the exact wording
- Use a shorter, more distinctive phrase
- Try the other approach (1 or 2)
- Consider whether the audio quality may be the issue (see below)

9. Difficult Audio Conditions

Recognition accuracy can be affected by: overlapping speech, multiple voices, background noise, music, sound effects, very low/high volume, distortion, strong accents, fast speech, unusual pronunciation, abbreviations, acronyms, echo/reverberation, poor-quality microphones, phone/radio-quality audio, compression artifacts, very short utterances, and sudden volume changes.

10. Tips for Better Search Results

- Use the wording you actually expect to be spoken.
- Prefer short but distinctive phrases over long, uncertain ones.
- If unsure of exact wording, try a shorter, more identifiable snippet instead of guessing a long sentence.

11. Command to run the program

```
python main.py
```

12. Running Tests (Optional, for Verification)

```
python -m pytest -v
```

This checks that key parts of the system work correctly: audio extraction, frame extraction, FPS/duration handling, text normalization, sentence matching, first-occurrence detection, not-found handling, progressive search, transcript reuse and persistence, timestamp ordering, and transcription behavior. Regular users don't need to run this.

13. Troubleshooting Guide

Problem	What to do
"python is not recognized"	Install Python and make sure it's added to PATH; verify with <code>python --version</code>
"ffmpeg/ffprobe not recognized"	Install FFmpeg and add it to PATH
"No module named faster_whisper"	Make sure your virtual environment is activated, then reinstall requirements
yt-dlp download fails	Check the URL is valid, check your internet connection, check if the source is supported, verify yt-dlp version
Download works but processing fails	Check the file in <code>data/raw/</code> for compatibility issues
Dialogue not found	See Section 10 above
Wrong occurrence returned	Use a longer, more distinctive phrase - the system intentionally returns the first match that qualifies

14. Full Walkthrough Example

```
cd <project path>
.venv\Scripts\Activate.ps1
python main.py
Enter video URL: <your video URL here>
Enter dialogue: but I'm very happy to be here
Enter choice (1/2): 2
```

Example progress output:

Transcribing 240.00s → 270.00s

Dialogue encountered.

Stopping transcription.

Example final result:

Timestamp: 00:05:24.840

Frame: 7788

Text: "My mind rebels at stagnation."

Frame saved:<path os the saved output frame>

15. Final Checklist Before Running

- Project folder is available on your computer
- Python is installed
- Virtual environment (.venv) is activated
- Dependencies are installed (requirements.txt)
- FFmpeg and FFprobe are available
- Internet connection is active
- You have a valid video URL
- You know the exact spoken dialogue you want to find
- Run `python main.py`